Geometry Name____________________
Unit 10 Date_____________________
Lesson 1 Homework Period____________________

Use the given context to answer questions 1-10.

Jamar orders an extra-large circular pizza that has an 18-inch (in.) diameter and is cut into 12 triangular slices.

1. What is the approximate area of an 18-inch pizza? Use 3.14 for π.

2. If the slices are cut precisely and equally, what fraction and percentage of the pizza is 6 slices?

3. What is the area of 6 slices?

4. What is the central angle of 6 slices?

5. If the slices are cut precisely and equally, what fraction of the pizza is 8 slices?

6. What is the area of 8 slices?

7. What is the central angle of 8 slices?

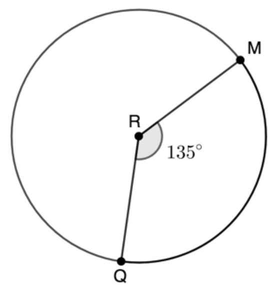
8. If the slices are cut precisely and equally, what fraction of the pizza is 1 slice?

9. What is the area of 1 slice?

10. What is the central angle of 1 slice?

Geometry Name____________________
Unit 10 Date_____________________
Lesson 2 Homework Period____________________

For questions 1-2, Circle R is shown where points M and Q are on the circle.

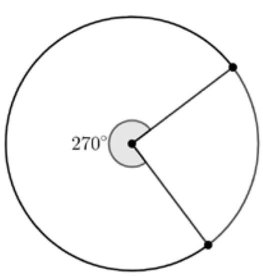


1. What is the ratio of the central angle, $\angle QRM$, to the whole circle? Simplify the ratio.

2. What fraction of the circle's circumference does $\widehat{MQ}$ represent?

For questions 3-8, find the ratio of the arc length to the circumference of the circle.

3.



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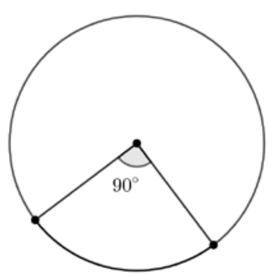

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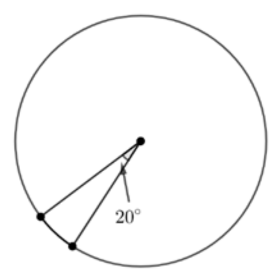

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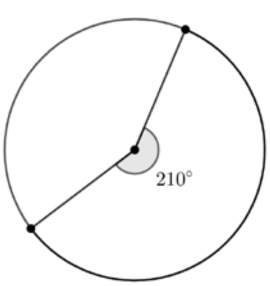

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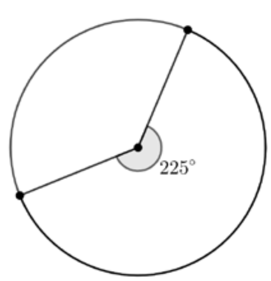

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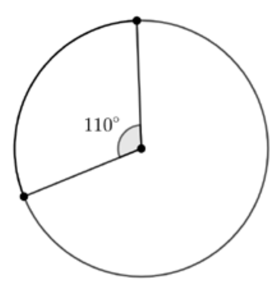
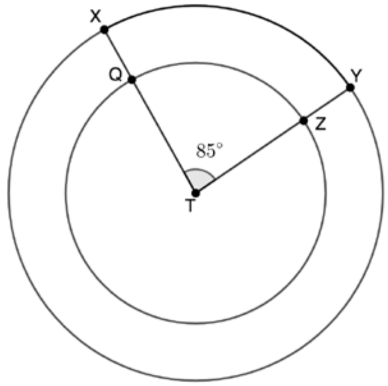


For questions 9-10, consider the following information.

Points X and Y lie on Circle T. The radius $\overline{\mathrm{TY}}$ measures 8 inches (in.). A smaller Circle T with a radius measure of 5 in. rests inside the larger Circle T, where points Q and Z lie on the smaller circle. The smaller Circle T has central angle $\angle QTZ$ that measures $85°$ and the larger Circle T has central angle $\angle XTY$ that measures $85°$.



9. What is the ratio of the arc length to circumference of Circle T with radius measure of 8 in.?

10. What is the ratio of the arc length to circumference of Circle T with radius measure of 5 in.?

Geometry
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Lesson 3 Homework

Name____________________
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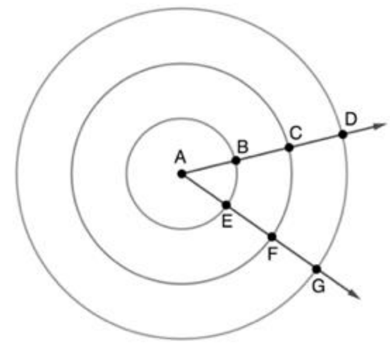
For questions 1-6, concentric circles with center A are shown. Points B and E lay on the innermost Circle A, and has radius $AB = 2$ units. Points C and F lay on the middle Circle A, and has radius $AC = 4$ units. Points D and G lay on the outer Circle A, and has radius $AD = 6$ units.



Determine the circumference for each circle. Leave your answer in terms of π.

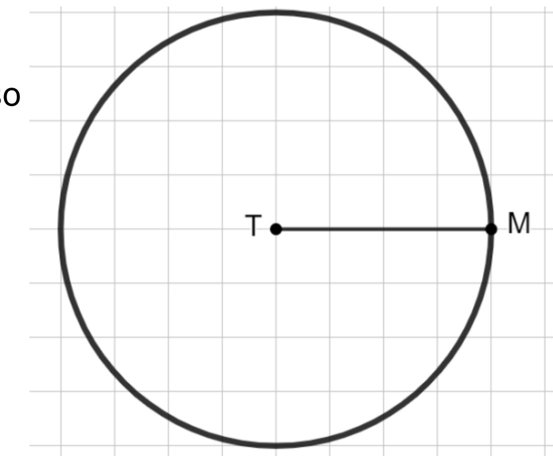
	Circle	Circumference
1.	Innermost Circle	
2.	Middle Circle	
3.	Outer Circle	

Complete the table.

	Circle	Ratio of Circumference to Radius
4.	Innermost Circle	
5.	Middle Circle	
6.	Outer Circle	

7. What would be the ratio of the circumference to the radius for a circle whose center is N and whose radius has a measurement of 9?

Circle T is shown where $\overline{TM}$ is a radius.



8. Label point R on the circumference of the circle so that the resulting arc has an approximate length of 1 radian.

9. Describe how you determined the location of point R.

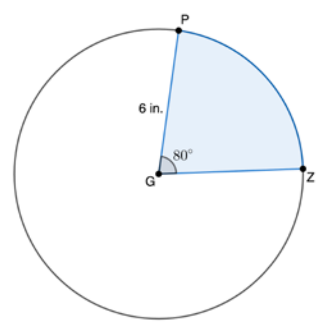
10. Label point W on the circumference of the circle so that the resulting arc has an approximate length of 2 radians.

Geometry Name____________________
Unit 10 Date_____________________
Lesson 4 Homework Period____________________

1. Circle B has a diameter of 14 inches (in.). Points H and W are on the circle and form the sector HBW. The central angle $\angle HBW$ in Circle B measures $150°$. Determine the area of the sector. Express your answer in terms of π.

2. The area of a sector with a radius of 10 centimeters (cm) is 148.28 square cm. Calculate the approximate measure of the central angle of the sector in degrees.

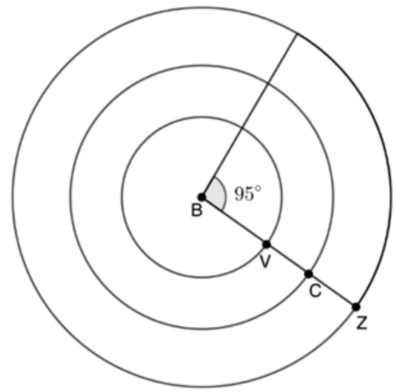
3. Circle G has radius $\overline{GP}$ that measures 6 inches (in.) with central angle $\angle PGZ$, where $m\angle PGZ = 80°$. Find the area of the shaded sector. Round your answer to the nearest hundredth.



4. Circle H has a radius of 9 inches (in.). Points M and N are on the circle and form the sector MHN. The central angle, $\angle MHN$, in Circle H measures $115°$. Determine the area of the sector. Express your answer in terms of π.

5. The area of a sector with a diameter of 12 feet (ft) is 62.8 square ft. Calculate the approximate measure of the central angle of the sector in degrees.

For questions 6-8, concentric circles with center B are shown with $m\angle B = 95°$. The smallest Circle B has a radius $\overline{BV}$ that measures 12 meters (m). The largest Circle B has radius $\overline{BZ}$ and the middle Circle B has radius $\overline{BC}$. Each circle has a radius that is 12 m longer than the immediate prior circle.

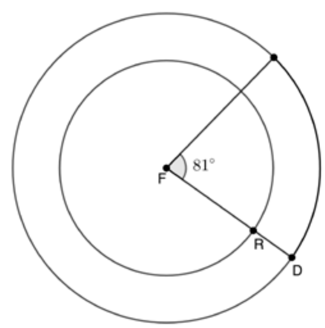


6. What is the arc length of the circle with radius $\overline{BV}$?

7. What is the arc length of the circle with radius $\overline{BC}$?

8. What is the arc length of the circle with radius $\overline{BZ}$?

For questions 9-10, concentric circles with center F are shown with $m\angle F = 81°$. The smallest Circle F has a radius $\overline{FR}$ that measures 15 meters (m) and the largest Circle F has radius $\overline{FD}$. The radius of the largest Circle F has a radius 7 m larger than the smaller Circle F.



9. What is the arc length of the circle with radius $\overline{FR}$?

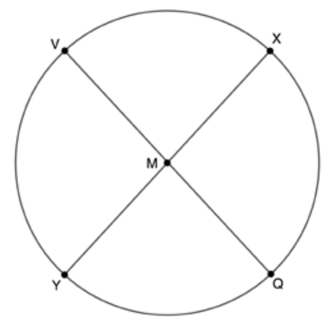
10. What is the arc length of the circle with radius $\overline{FD}$?

Geometry Name____________________
Unit 10 Date_____________________
Lesson 5 Homework Period___________________

1. Circle Y has points X, M, and R on the circle with $m\angle MYX = 37.2°$. If $m\widehat{MR} = 146.8°$ and $\widehat{MR}$ does not intersect $\widehat{MX}$, what is $m\angle RYX$?

2. Circle G has radii $\overline{GH}$ and $\overline{GK}$. Find the value of x if $m\angle KGH = 122°$ and $m\widehat{KH} = (5x + 7)°$.

For questions 3-4, Circle M has diameters $\overline{VQ}$ and $\overline{XY}$ that both intersect at point M.



3. Given $m\widehat{QY} = 80°$, find the $m\widehat{VX}$.

4. Given $m\widehat{QX} = 100°$, find the $m\widehat{YV}$.

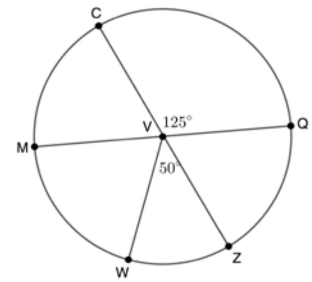
For questions 5-6, consider the following context.

Lorenzo is creating a folding fan. He needs to determine the measure of the arc of the fan before he can cut the paper. He uses the expression $(5x + 119)°$ as the measure of the central angle, $\angle JQZ$. For the $m\widehat{ZJ}$, he uses the expression $(25x + 59)°$.

5. What is the value of x?

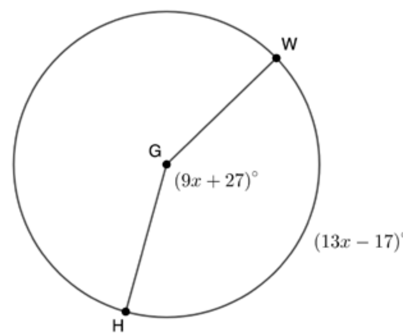
6. What is the degree measure of the arc of the fan, $\widehat{ZJ}$?

7. In Circle V, $m\angle CVQ = 125°$ and $m\angle ZVW = 50°$. Calculate the $m\widehat{CQW}$, given that V is the center and $\overline{CZ}$ and $\overline{MQ}$ are diameters.



8. Circle Y has points X, Q, and W on the circle. The measure of the central angle, $\angle WYQ = 65°, m\widehat{XW} = 117°$, and $\widehat{XW}$ does not intersect $\widehat{WQ}$. Find $m\angle XYQ$.

For questions 9-10, consider the following context.



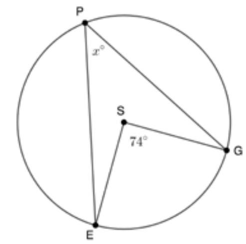
A horse is tied to a post at point G and is grazing in a circular pattern. She decides to walk from her position at point H to the trough of water at point W along the circumference of the circular area she can roam. The measure of the central angle, $\angle WGH$, is $(9x + 27)°$, and the measure of the arc that the horse walked, $\widehat{WH}$, is $(13x - 17)°$.

9. Find the value of x.

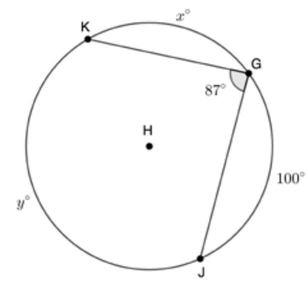
10. Find the measure of the arc that the horse walked, $\widehat{WH}$.

Geometry Name____________________
Unit 10 Date_____________________
Lesson 6 Homework Period___________________

1. Circle S is shown where $\angle GSE$ is a central angle, $\angle GPE$ is inscribed in the circle, and $m\angle GSE = 74°$. Find $m\angle GPE$.



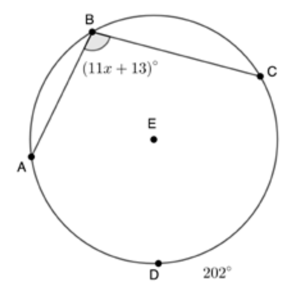
For questions 2-3, Circle H has points K, G, and J on the circle. $\angle KGJ$ is inscribed in Circle H measuring $87°$, and the $m\widehat{GJ} = 100°$.



2. Find the $m\widehat{KJ}$.

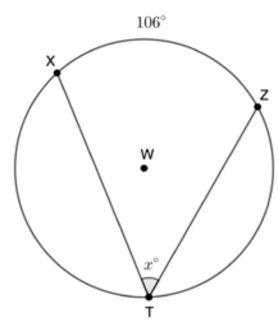
3. Find the $m\widehat{KG}$.

For questions 4-5, Points A, B, C, and D lie on Circle E. $\angle ABC$ is inscribed in the circle, the $m\angle ABC = (11x + 13)°$, and the $m\widehat{CDA} = 202°$.



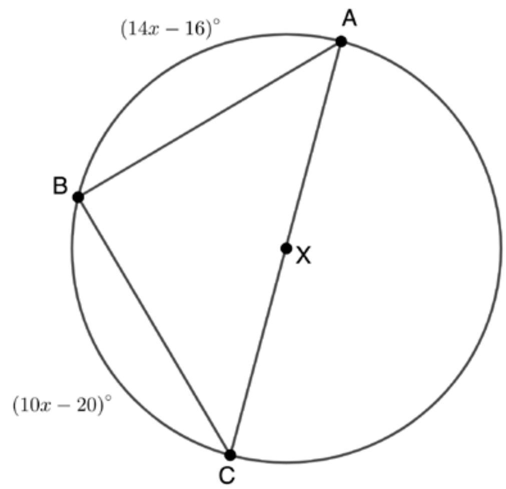
4. Find the $m\angle ABC$.

5. Find the value of x.



6. Circle W is shown, where points X, Z, and T are on the circle. $\angle XTZ$ is inscribed in Circle W with $m\widehat{XZ} = 106°$. Find the value of x.

For questions 7-10, Circle X is shown with inscribed angle $\angle ABC$ and diameter $\overline{CA}$. The $m\widehat{AB} = (14x - 16)°$ and the $m\widehat{BC} = (10x - 20)°$.



7. Find the value of x.

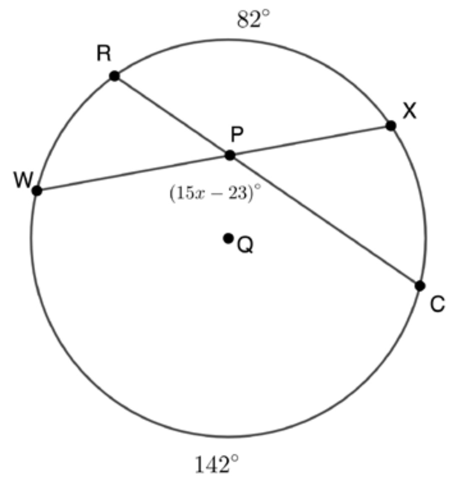
8. Find the $m\angle BAC$.

9. Find the $m\angle ACB$.

10. Find the $m\widehat{CB}$.

Geometry | Name____________________
Unit 10 | Date_____________________
Lesson 7 Homework | Period___________________

For questions 1-3, Circle Q is shown, where chords $\overline{RC}$ and $\overline{WX}$ intersect at point P, $m\widehat{CW} = 142°, m\widehat{RX} = 82°,$ and $m\angle WPC = (15x - 23)°$.

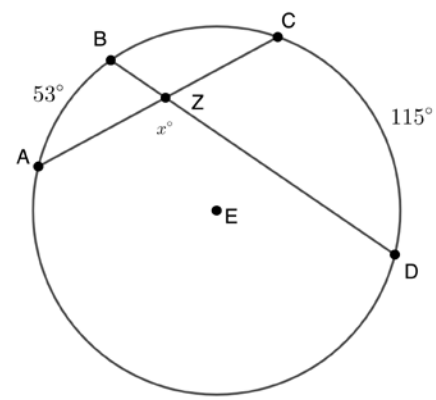


1. Find the $m\angle WPC$.

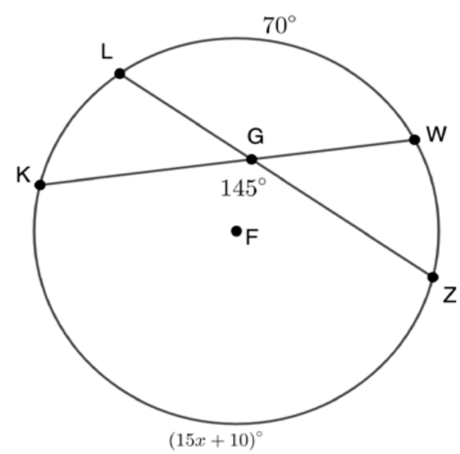
2. Write a formula to find the value of x.

3. Find the value of x.

4. Circle E is shown, where chords $\overline{BD}$ and $\overline{CA}$ intersect at point Z. Find the value of x for $\angle AZD$, given $m\widehat{CD} = 115°$ and $m\widehat{AB} = 53°$.



For questions 5-7, Circle F is shown, where chords $\overline{LZ}$ and $\overline{KW}$ intersect at point G, $m\widehat{LW} = 70°$, $m\widehat{KZ} = (15x + 10)°$, and $m\angle KGZ = 145°$.

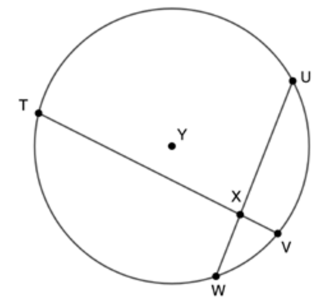


5. Write a formula to find the value of x.

6. Find the value of x.

7. Find the $m\widehat{KZ}$.

For questions 8-10, Circle Y is shown, where chords $\overline{VT}$ and $\overline{UW}$ intersect at point X.



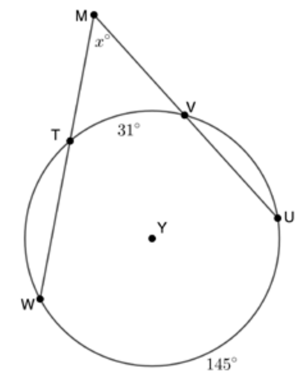
8. If $m\widehat{TU} = 128°$ and $m\widehat{VW} = 30°$, what is the measure of $\angle TXU$?

9. If $m\widehat{TW} = 152°$ and $m\widehat{UV} = 40°$, what is the measure of $\angle TXW$?

10. If $m\widehat{VW} = 56°$ and $m\angle TXU = 78°$, what is the measure of $\widehat{TU}$?

Geometry — Name____________________
Unit 10 — Date_____________________
Lesson 8 Homework — Period___________________

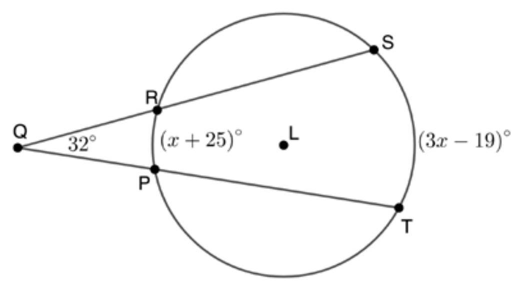
For questions 1-2, Circle Y has points W, T, V, and U on the circle. Secant lines $\overline{WM}$ and $\overline{UM}$ intersect at point M outside the circle. The $m\widehat{UW} = 145°$, $m\widehat{TV} = 31°$, and $m\angle TMV = x°$.



1. Write a formula that can be used to find the value of x.

2. Find the value of x.

For questions 3-5, Circle L has points T, P, R, and S on the circle. Secant lines $\overline{TQ}$ and $\overline{SQ}$ intersect at point Q outside the circle. The $m\widehat{ST} = (3x - 19)°$, $m\widehat{PR} = (x + 25)°$, and $m\angle PQR = 32°$.

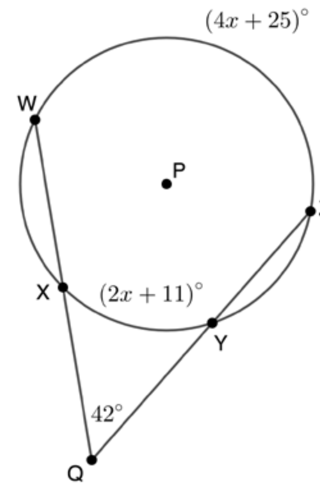


3. Find the value of x.

4. Find the $m\widehat{PR}$.

5. Find the $m\widehat{ST}$.

For questions 6-8, Circle P has points W, X, Y, and Z on the circle. Secant lines $\overline{WQ}$ and $\overline{ZQ}$ intersect at point Q outside the circle. The $m\widehat{WZ} = (4x + 25)°$, $m\widehat{XY} = (2x + 11)°$, and $m\angle XQY = 42°$.

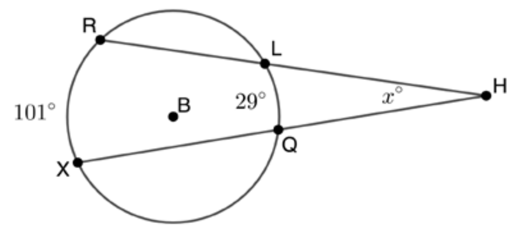


6. Find the value of x.

7. Find the $m\widehat{XY}$.

8. Find the $m\widehat{WZ}$.

For questions 9-10, Circle B has points R, L, Q, and X on the circle. Secant lines $\overline{XH}$ and $\overline{RH}$ intersect at point H outside the circle. The $m\widehat{RX} = 101°$, $m\widehat{LQ} = 29°$, and $m\angle LHQ = x°$.

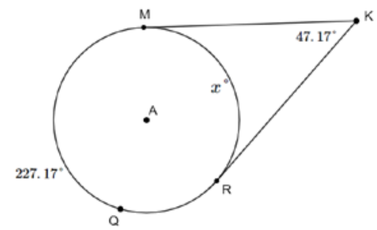


9. Write a formula that can be used to find the value of x.

10. Find the value of x.

Geometry Name____________________
Unit 10 Date_____________________
Lesson 9 Homework Period____________________

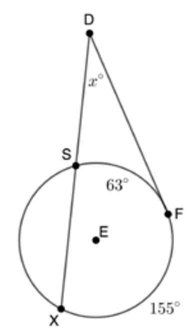
For questions 1-2, Circle A has points $M, Q,$ and R on the circle. Tangents $\overline{MK}$ and $\overline{RK}$ intersect at point K outside the circle. The $m\widehat{MQR} = 227.17°, m\widehat{MR} = x°$, and $m\angle MKR = 47.17°$.



1. Write a formula that can be used to find the value of x.

2. Find the value of x.

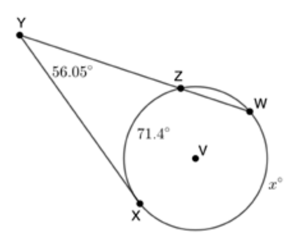
For questions 3-4, Circle E has points X, S, and F on the circle. Tangent $\overline{FD}$ and secant $\overline{XD}$ intersect at point D outside the circle. The $m\widehat{XF} = 155°$, $m\widehat{SF} = 63°$, and $m\angle SDF = x°$.



3. Write a formula that can be used to find the value of x.

4. Find the value of x.

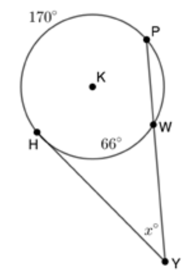
For questions 5-6, Circle V has points X, Z, and W on the circle. Tangent $\overline{XY}$ and secant $\overline{WY}$ intersect at point Y outside the circle. The $m\widehat{XW} = x°$, $m\widehat{XZ} = 71.4°$, and $m\angle XYZ = 56.05°$.



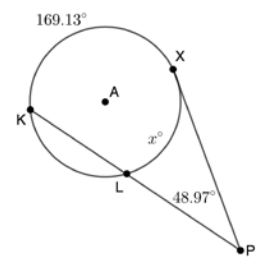
5. Write a formula that can be used to find the value of x.

6. Find the value of x.

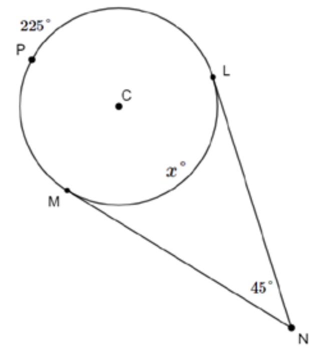
7. Circle K has points P, W, and H on the circle. Tangent $\overline{HY}$ and secant $\overline{PY}$ intersect at point Y outside the circle. The $m\widehat{HP} = 170°$, $m\widehat{HW} = 66°$, and $m\angle HYW = x°$. What is the $m\angle HYW$?



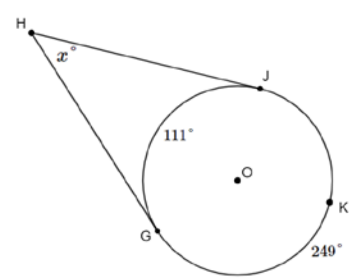
8. Circle A has points X, L, and K on the circle. Tangent $\overline{XP}$ and secant $\overline{KP}$ intersect at point P outside the circle. The $m\widehat{KX} = 169.13°$, $m\widehat{XL} = x°$, and $m\angle XPL = 48.97°$. What is the $m\widehat{XL}$?



9. Circle C has points L, P, and M on the circle. Tangents $\overline{LN}$ and $\overline{MN}$ intersect at point N outside the circle. The $m\widehat{LPM} = 225°$, $m\widehat{LM} = x°$, and $m\angle LNM = 45°$. What is the $m\widehat{LM}$?

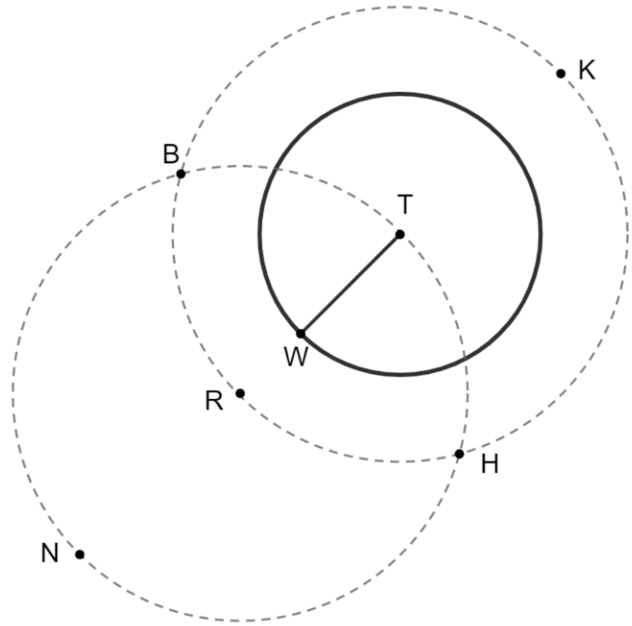


10. Circle O has points G, J, and K on the circle. Tangents $\overline{GH}$ and $\overline{JH}$ intersect at point H outside the circle. The $m\widehat{GKJ} = 249°$, $m\widehat{GJ} = 111°$, and $m\angle GHJ = x°$. What is the $m\angle GHJ$?



Geometry Name____________________
Unit 10 Date_____________________
Lesson H1 Homework Period___________________

Circle T is shown with radius $\overline{TW}$. The two dotted circles were constructed using a compass so that they are congruent to each other. Points K, H, R, and B are on the dotted Circle T and points B, H, T, and N on the dotted Circle R.



1. Select all of the line segments that are congruent to each other.

- □ $\overline{BK}$
- □ $\overline{BR}$
- □ $\overline{BT}$
- □ $\overline{TH}$
- □ $\overline{TK}$
- □ $\overline{TR}$
- □ $\overline{TW}$

2. Justify why the line segments chosen in question 1 are congruent.

3. Select all of the line segments that are congruent to each other.

- □ $\overline{BH}$
- □ $\overline{BN}$
- □ $\overline{BR}$
- □ $\overline{NH}$
- □ $\overline{NR}$
- □ $\overline{RH}$
- □ $\overline{RT}$
- □ $\overline{RW}$

4. Justify why the line segments chosen in question 3 are congruent.

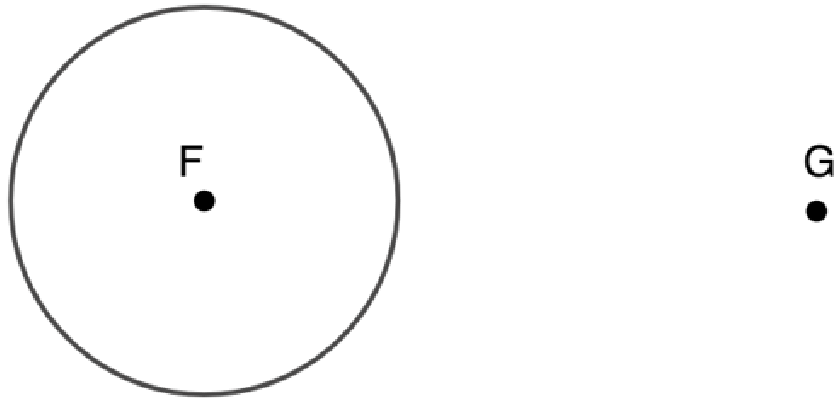
5. Draw $\overleftrightarrow{NK}$ on the diagram.

6. Draw $\overleftrightarrow{BH}$ on the diagram.

7. Describe how $\overleftrightarrow{NK}$ and $\overleftrightarrow{BH}$ are related.

8. Justify whether $\overleftrightarrow{BH}$ is tangent to Circle T.

Circle F and point G are shown.



9. Construct the perpendicular bisector of $\overline{FG}$.

10. Construct a line that is tangent to Circle F and goes through point G.